

overlap-calculator Case Studies

Index of all 11 worked examples

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1 overlap-calculator Case Studies

This directory contains 11 worked case studies, each isolating one **user-selectable feature** of **overlap-calculator**. Every case study ships in three documentation formats — Markdown, LaTeX source, and a typeset PDF — so the same content is browsable on GitHub, editable as LaTeX, and printable as a stand-alone PDF.

Each case study is intentionally small (**2 samples** $\times$ **2 light sources**) so a full run finishes in seconds and the repository stays lean. The case isolates one knob at a time: light source, plot DPI, broadening width, Beer–Lambert reference, wavelength grid, sheet selection, etc.

1.1 Layout shared by every case

```
case_studies/<NN>_<name>/
+-- input.json
+-- submit.py
+-- README.md
+-- README.tex
+-- README.pdf
+-- output/
    +-- cli/
    |   +-- tables/{results,results_timings,descriptor_summary}.*
    |   +-- plots/...
    +-- api/
        +-- analysis_outputs.zip
        +-- tables/...
        +-- plots/...
```

`input.json` is the 2-entry manifest that points at the bundled files under `../input/files/`. `submit.py` drives both the CLI and the HTTP API in one run. The Markdown README is the narrative for the case; the LaTeX and PDF carry the same content in typeset form. The `output/` tree is produced once you run `submit.py`; case study 07 produces no `plots/` directory because it turns plotting off.

1.2 Case study index

If this is your first time using `overlap-calculator`, start with **Case Study 01** — its **§1 Prerequisites** is the install / environment / API-startup walkthrough that every other case study links to instead of repeating.

Case	Folder	Feature exercised	Inputs	LaTeX	PDF
01	01_theoretical_only	Canonical TD-DFT-only workflow (anchor)	2 <code>.out</code>	tex	pdf
02	02_experimental_only	Measured spectra only	2 series in 1 <code>.xlsx</code>	tex	pdf
03	03_mixed_theory_experiment	Theoretical + experimental in one manifest	2 <code>.out</code> + 2 series	tex	pdf
04	04_custom_light_source	<code>--light-source-file</code> / <code>light_source_files</code>	2 <code>.out</code> + 1 CSV	tex	pdf
05	05_default_light_source_selection	<code>--default-light-sources</code> subset	2 <code>.out</code>	tex	pdf
06	06_plot_dpi	<code>--plot-dpi</code> 600 for publication TIFFs	2 <code>.out</code>	tex	pdf
07	07_table_only	<code>--no-plot-outputs</code> / <code>plot_outputs=false</code>	2 <code>.out</code>	tex	pdf
08	08_broadening_sigma	<code>--sigma-ev</code> (0.30 $\rightarrow$ 0.20 eV)	2 <code>.out</code>	tex	pdf
09	09_beer_lambert_tuning	<code>--concentration-m</code> and <code>--path-cm</code>	2 <code>.out</code>	tex	pdf
10	10_wavelength_grid	<code>--wl-min</code> / <code>--wl-max</code> / <code>--num-points</code>	2 <code>.out</code>	tex	pdf
11	11_sheet_overrides	<code>sheet_name</code> (CLI) / <code>sheet_overrides</code> (API)	2 series in 1 <code>.xlsx</code>	tex	pdf

1.3 Bundled inputs

Every case study draws from the same files in `../input/files/`:

- 13 Gaussian TD-DFT outputs (the case studies use 2 of them to keep runs short).
- One PhotochemCAD-derived Excel workbook with 8 numeric absorbance series (the case studies use 2 of them).

Citation for the experimental workbook. The bundled experimental absorbance data is derived from M. Taniguchi & J. S. Lindsey, "Database of absorption and fluorescence spectra of >300 common compounds for use in PhotochemCAD", *Photochem. Photobiol.* **94** (2018) 290–327. See the project [README](#) for the full acknowledgement and BibTeX entries (covering both the bundled light sources and the experimental workbook).